

Case Study: Calculating Exponents

Although recursion doesn't necessarily produce better code, taking a recursive approach can give you new insights into your programming problem. As a case study, let's examine how to calculate exponents.

Exponents are calculated by multiplying a number by itself. For example, the exponent “three raised to the sixth power,” or 3^6 , is equal to multiplying 3 by itself six times: $3 \times 3 \times 3 \times 3 \times 3 \times 3 = 729$. This is such a common operation that Python has the `**` operator and JavaScript has the built-in `Math.pow()` function to perform exponentiation. We can calculate 3^6 with the Python code `3 ** 6` and with the JavaScript code `Math.pow(3, 6)`.

But let's write our own exponent-calculating code. The solution is straightforward: create a loop that repeatedly multiplies a number by itself and returns the final product. Here is an iterative *exponentByIteration.py* Python program:

Python

```
def exponentByIteration(a, n):
    result = 1
    for i in range(n):
        result *= a
    return result

print(exponentByIteration(3, 6))
print(exponentByIteration(10, 3))
print(exponentByIteration(17, 10))
```

And here is an equivalent JavaScript *exponentByIteration.html* program:

JavaScript

```
<script type="text/javascript">
function exponentByIteration(a, n) {
    let result = 1;
    for (let i = 0; i < n; i++) {
        result *= a;
    }
}
```
